**HW1 Math 638 Differential Geometry II, Winter 2026**

Due on class in paper, Wednesday January 14, 2026.

**Q1.1**. ([Lee, 2nd, 7-2, p.171]) Let $G$ be a Lie group.

(1a) Let $m : G \times G \to G$ denote the multiplication map. Using [Lee, 2ed, Proposition 3.14] to identify $T_{(e,e)} : (G \times G)$ with $T_eG \oplus T_eG$, show that the differential $dm_{(e,e)} : T_eG \oplus T_eG \to T_eG$ is given by

$$dm_{(e,e)}(X, Y) = X + Y.$$

[Hint: compute $dm_{(e,e)}(X, 0)$ and $dm_{(e,e)}(0, Y)$ separately.]

(1b) Let $i : G \to G$ denote the inversion map. Show that $di_e : T_eG \to T_eG$ is given by $di_e(X) = -X$.

**Q1.2**. ([Lee, 2nd, 7-3, p.171]) Show that smoothness of the inversion map is redundant in the definition of Lie group: if $G$ is a smooth manifold with a group structure such that the multiplication map $m : G \times G \to G$ is smooth, then $G$ is a Lie group.

[Hint: show that the map $F : G \times G \to G \times G$ defined by $F(g, h) = (g, gh)$ is a bijective local diffeomorphism.]

**Q1.3**. ([Lee, 2nd, 7-4, p.171]) Let $\det : GL(n; \mathbb{R}) \to \mathbb{R}$ denote the determinant function. Use [Lee, 2ed, Corollary 3.25] to compute the differential of det, as follows.

(3a) For any $A \in M(n; \mathbb{R})$, show that

$$\left.\frac{d}{dt}\right|_{t=0} \det\left(I_n + tA\right) = \operatorname{tr} A,$$

where $\operatorname{tr} A$ is the trace of matrix $A$.

(3b) For $X \in GL(n; \mathbb{R})$ and $B \in T_X GL(n; \mathbb{R}) \simeq M(n; \mathbb{R})$ (the space of real $n \times n$ matrices), show that

$$d(\det)_X(B) = (\det X) \operatorname{tr}(X^{-1}B).$$

[Hint: $\det(X + tB) = (\det X) \cdot \det(I_n + tX^{-1}B)$.]

**Q1.4**. ([Lee, 2nd, 7-10, p.172]) Show that the formula

$$A \cdot [\vec{z}] = [A\vec{z}]$$

defines a smooth, transitive left action of $GL(n + 1; \mathbb{C})$ on $\mathbb{C}\mathrm{P}^n$.

**Q1.5**. ([Lee, 2nd, 7-11, p.172]) Considering $S^{2n+1}$ as the unit sphere in $\mathbb{C}^{n+1}$, define an action of $S^1$ on $S^{2n+1}$, called the Hopf action, by

$$z \cdot (w^1, \cdots, w^{n+1}) = (zw^1, \cdots, zw^{n+1}).$$

Show that this action is smooth and its orbits are disjoint unit circles in $\mathbb{C}^{n+1}$ whose union is $S^{2n+1}$.

**Q1.6**. ([Lee, 2nd, 7-13 and 14, p.172]) (6a) For each $n \geq 1$, prove that $U(n)$ is a properly embedded real $n^2$-dimensional Lie subgroup of $GL(n;\mathbb{C})$.

(6b) For each $n \geq 1$, prove that $SU(n)$ is a properly embedded $(n^2 - 1)$-dimensional Lie subgroup of $U(n)$.